

RODINO, Russian Federation

WMO: 360200

Lat: 52.486N

Lon: 80.183E

Elev: 520

StdP: 14.42

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-28.5	-38.5	0.6	-33.4	-33.0	0.9	-28.4	29.8	24.0	27.1	20.6	2.1	200	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	88.3	65.2	85.3	64.2	82.1	63.4	68.6	81.8	67.1	79.8	65.7	77.8	8.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	91.7	72.9	62.7	86.8	71.7	61.2	82.2	71.2	33.2	81.8	32.0	79.7	30.8	77.8	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	22.6	19.9	-35.2	92.8	8.1	4.6	-41.0	96.1	-45.7	98.8	-50.3	101.4	-56.2	104.8		
			-35.4	71.8	8.0	2.0	-41.1	73.2	-45.8	74.4	-50.3	75.5	-56.1	76.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	0.3	3.8	21.2	43.8	54.4	67.0	68.8	64.0	52.7	38.9	22.2	8.9
	DBStd	26.52	16.09	14.11	12.07	10.60	9.31	7.63	5.86	6.37	8.61	9.07	14.16	14.87
	HDD50	6569	1542	1295	891	240	63	2	0	0	66	362	834	1274
	HDD65	10467	2007	1715	1356	638	345	68	30	94	381	810	1284	1739
	CDD50	1946	0	0	0	53	200	512	583	435	147	16	0	0
	CDD65	371	0	0	0	1	17	129	148	64	12	0	0	0
	CDH74	3729	0	0	0	23	224	1203	1309	749	221	0	0	0
	CDH80	1120	0	0	0	2	42	387	411	202	76	0	0	0
Wind	WSAvg	8.2	8.3	8.2	8.9	9.2	9.5	7.3	6.4	6.7	7.2	7.7	9.3	9.8
Precipitation	PrecAvg	13.0	0.6	0.5	0.7	0.8	1.0	1.5	2.3	1.5	1.0	1.1	1.1	0.9
	PrecMax	16.8	1.1	1.1	1.6	1.9	3.1	2.8	4.3	3.8	2.1	3.0	1.9	1.5
	PrecMin	7.2	0.2	0.0	0.2	0.0	0.3	0.3	0.6	0.1	0.1	0.4	0.4	0.4
	PrecStd	2.1	0.3	0.3	0.4	0.5	0.6	0.7	1.1	0.7	0.6	0.6	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	34.1	51.8	77.6	85.4	91.3	92.5	91.0	88.3	70.8	52.0	35.7
		MCWB	32.0	31.9	43.6	56.7	60.6	65.8	68.3	66.1	63.1	51.3	46.1	33.9
	2%	DB	30.3	30.6	43.5	71.4	79.9	87.7	88.0	85.3	81.0	63.2	46.4	33.3
		MCWB	28.7	28.6	37.6	53.2	57.9	64.5	67.1	64.1	58.9	50.2	41.9	31.9
	5%	DB	25.3	27.6	39.2	65.9	76.0	84.6	84.6	81.1	74.1	57.6	41.9	30.9
		MCWB	23.8	25.9	35.6	50.5	56.4	63.0	66.2	62.4	56.5	47.4	38.0	29.4
	10%	DB	22.1	24.1	36.1	60.6	71.6	81.2	81.3	78.0	68.6	52.3	38.5	28.1
		MCWB	20.8	22.5	33.3	48.2	54.7	63.0	65.3	61.7	54.1	45.3	35.6	26.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	32.3	44.0	58.2	63.0	68.9	71.0	68.9	66.6	54.1	47.0	34.1
		MCDB	34.8	33.8	51.3	74.7	77.1	81.6	85.3	85.1	85.5	63.2	51.0	35.3
	2%	WB	28.7	28.9	38.1	54.9	60.2	67.3	69.4	66.1	60.6	51.5	42.0	32.1
		MCDB	30.3	30.7	42.4	68.6	75.4	80.6	82.7	79.9	74.0	61.0	45.9	33.2
	5%	WB	24.1	25.9	35.8	51.8	58.0	65.9	68.1	64.5	58.0	48.3	38.1	29.4
		MCDB	25.3	27.5	38.9	63.1	72.0	79.5	80.2	77.4	71.5	57.8	41.5	30.6
	10%	WB	20.7	22.5	33.5	49.2	55.8	64.6	66.6	62.9	55.1	45.7	35.3	26.8
		MCDB	22.0	24.0	36.1	59.8	69.7	78.0	78.1	75.2	67.1	51.7	38.1	28.1
Mean Daily Temperature Range	5% DB	MDBR	12.6	14.8	14.6	18.9	21.2	19.4	18.5	21.0	19.7	13.4	10.4	13.0
		MCDBR	12.2	11.9	15.3	26.5	26.3	24.4	24.0	26.9	29.2	21.7	12.9	11.8
		MCWBR	12.1	11.4	11.1	13.7	11.7	8.0	8.4	11.0	12.8	12.1	9.4	12.0
	5% WB	MCDBR	12.3	12.1	14.2	23.2	22.3	19.3	19.5	22.6	24.6	19.3	12.1	11.4
		MCWBR	12.3	11.8	10.6	13.3	11.8	8.1	8.0	10.1	12.4	11.6	9.2	11.7
Clear-Sky Solar Irradiance	taub	0.300	0.336	0.389	0.410	0.414	0.414	0.438	0.417	0.375	0.354	0.339	0.308	
	taud	2.022	1.949	1.915	2.141	2.181	2.269	2.200	2.225	2.288	2.244	2.087	1.980	
	Ebn at Noon	204	232	244	260	265	266	257	255	252	229	187	172	
	Edh at Noon	25	37	48	43	43	40	43	39	33	28	24	23	
All-Sky Solar Radiation	RadAvg	384	736	1201	1500	1885	2015	1829	1593	1160	621	312	261	
	RadStd	30	38	99	108	138	134	123	105	98	86	39	38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.88	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page